

CS415 Assignment 1

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I. QUESTION 1

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. The convolution $f * g$ is defined for each $x \in \mathbb{R}$ by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy.$$

We prove that

$$(f * g)(x) = (g * f)(x) \quad \text{for all } x \in \mathbb{R}.$$

The minus sign from $du = -dy$ is canceled when reversing integration limits. The same idea extends to convolution.

Description Of The Answer:

We have two real-valued functions f and g . Their convolution is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy,$$

The goal of this question is to show that convolution is *commutative*.

$$(f * g)(x) = (g * f)(x) \quad \text{for every } x.$$

We assume f and g are integrable. The proof above is short. Make the simple substitution $u = x - y$, and the sign from $du = -dy$ disappears when we flip the integration limits. The integral we get is the definition of $(g * f)(x)$. It holds for every x , we conclude $f * g = g * f$.

Math Proof: $u = x - y$ (so $du = -dy$), we obtain

$$\begin{aligned} (f * g)(x) &= \int_{-\infty}^{\infty} f(y) g(x - y) dy \\ &= \int_{+\infty}^{-\infty} f(x - u) g(u) (-du) \\ &= \int_{-\infty}^{\infty} f(x - u) g(u) du \\ &= \int_{-\infty}^{\infty} g(u) f(x - u) du = (g * f)(x). \end{aligned}$$

II. QUESTION 2

$$(f \star g)(x) = \int_{-\infty}^{+\infty} f(y) g(x+y) dy.$$

Claim: Cross-correlation is not generally commutative: there exist f, g and x with $(f \star g)(x) \neq (g \star f)(x)$.

Suppose f is even and g is odd: $f(-u) = f(u)$ and $g(-u) = -g(u)$. Then, for all $x \in \mathbb{R}$,

$$(g \star f)(x) = -(f \star g)(x).$$

Description of the Answer:

Functions f and g , their cross-correlation is defined as

$$(f \star g)(x) = \int_{-\infty}^{+\infty} f(y) g(x+y) dy.$$

The goal of this question is to show that the cross-correlation is *not* commutative in general; that is, there exist functions for which $(f \star g)(x) \neq (g \star f)(x)$.

The key idea is that swapping f and g . We can achieve this by picking one function that is even and another that is odd. Then a short changed variable shows the two correlations differ by a minus sign. This proves the order matters for correlation, so in general $(f \star g) \neq (g \star f)$.

Math Proof:

$$\begin{aligned} (g \star f)(x) &= \int_{-\infty}^{+\infty} g(y) f(x+y) dy \\ &= \int_{-\infty}^{+\infty} g(-x-u) f(-u) (-du) \quad (u = -x-y, dy = -du) \\ &= \int_{-\infty}^{+\infty} g(-x-u) f(-u) du \\ &= \int_{-\infty}^{+\infty} (-g(x+u)) f(u) du \quad (f(-u) = f(u), g(-x-u) = -g(x+u)) \\ &= - \int_{-\infty}^{+\infty} f(u) g(x+u) du = -(f \star g)(x). \end{aligned}$$

This Gives:

$$(g \star f)(x) \neq -(f \star g)(x).$$

III. QUESTION 3: RELATED COLAB NOTEBOOK

The link of the Notebook: [Click for Google Colab notebook.](#)

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